FAULT CODE 2727 - SAE J1939 Data Link 2 Engine Network - Abnormal Update Rate

Warnings and Cautions

⚠ CAUTION ⚠

To reduce the possibility of pin and harness damage, use the following test leads when taking a measurement: Part Number 3822758 - male Deutsch™/AMP™/Metri-Pack™ test lead, Part Number 3822917 - female Deutsch™/AMP™/Metri-Pack™ test lead, and Part Number 3823995 - male Weather Pack™ test lead.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Check the fault codes.	
	STEP 1A. Check for an active fault code.	Fault Code 2727 active?
STEP 2.	Check the original equipment manufacturer (OEM) power harness and engine harness.	
	STEP 2A. Inspect the 4-pin power connector and the ECM connector pins.	Dirty or damaged pins?
	STEP 2B. Check for an open circuit in the battery power circuits.	Voltage between 10.0 to 15.0-VDC for 12 volt system or 21 to 27-VDC for 24 volt system?
	STEP 2C. Check the battery supply for a short circuit from pin-to-pin.	Greater than 100k ohms?
	STEP 2D. Check the add-on or accessory wiring at the positive (+) battery terminal.	Damaged wires?
	STEP 2E. Check the ignition input-to-ECM wire.	Ignition wire uninterrupted?

STEPS		SPECIFICATIONS
	STEP 2F. Check the ignition input circuit.	Less than 5 ohms?
STEP 3.	Check the Society of Automotive Engineering (SAE) J1939 data link 2 circuit.	
	STEP 3A. Inspect all engine harness connector pins.	Dirty or damaged pins?
	STEP 3B. Check the SAE J1939 data link 2 engine network resistance in the engine harness.	Between 54 to 66 ohms?
	STEP 3C. Check the SAE J1939 data link 2 engine network terminator resistor.	Between 108 to 132 ohms?
	STEP 3D. Check for ECM communication to INSITE™ electronic service tool.	Does the ECM communicate?
STEP 4.	Clear the fault codes.	
	STEP 4A. Disable the fault code.	Fault Code 2727 inactive?
	STEP 4B. Clear the inactive fault codes.	All fault codes cleared?

STEP 1. Check the fault codes.

STEP 1A. Check for an active fault code.

Conditions :

- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step

Check for an active fault code. • Use INSITE™ electronic service tool to read the fault codes.	Fault Code 2727 active? **YES**	**2A**
	Fault Code 2727 active? **NO**	**Use the following procedure for an inactive or intermittent fault code. Refer to Procedure 019-362 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)**

STEP 2. Check the OEM power harness and engine harness.

STEP 2A. Inspect the 4-pin power connector and the ECM connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the 4-pin power connector from the ECM.

Action	Specification/Repair	Next Step

Inspect the ECM harness connector and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris on or in the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? **YES** **Repair :** Clean the connector and pins. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. • Repair the engine harness. Refer to Procedure 019-043 in Section 19 (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html). • Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	**4A**
	Dirty or damaged pins? **NO**	**2B**

STEP 2B. Check for an open circuit in the battery power circuits.

Conditions :

- Reference the circuit diagram or wiring diagram for connector pin for identification.

Action	**Specification/Repair**	**Next Step**

<table>
<tr>
<td rowspan="2">Check for an open circuit in the battery power circuits.
Measure the voltage between the battery voltage SUPPLY pin in the 4-pin power connector to engine block ground.
Reference the circuit diagram or wiring diagram for connector pin for identification.</td>
<td>Voltage between 10.0 to 15.0 VDC for 12 volt system or 21 to 27 VDC for 24 volt system?

YES</td>
<td>2C</td>
</tr>
<tr>
<td>Voltage between 10.0 to 15.0 VDC for 12 volt system or 21 to 27 VDC for 24 volt system?

NO

Repair :

Troubleshoot all harnesses connected in series to determine which contains the open circuit.

Refer to the circuit diagram or wiring diagram for all harness interconnections.

Replace the damaged section of the harness.
Repair the engine harness. Refer to Procedure 019-043 in Section 19 (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html).
Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)</td>
<td>4A</td>
</tr>
</table>

STEP 2C. Check the battery supply for a short circuit from pin to pin.

Conditions :

- Turn keyswitch OFF.
- Disconnect the 4-pin power harness connector from the ECM connector.
- Disconnect the battery leads.

<table>
<tr><th>Action</th><th>Specification/Repair</th><th>Next Step</th></tr>
<tr><td rowspan="2">Check for a short circuit from pin to pin.
Measure the resistance from a battery voltage SUPPLY pin in the 4-pin power connector to all other pins in the connector.
Measure the resistance from a battery voltage RETURN pin in the 4-pin power connector to all other pins in the connector.
Refer to the circuit diagram or wiring diagram for connector pin identification.

Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)</td><td>Greater than 100k ohms?

YES</td><td>2D</td></tr>
<tr><td>Greater than 100k ohms?

NO

Repair :

Troubleshoot all harnesses connected in series to determine which contains the short circuit.

Refer to the circuit diagram or wiring diagram for all harness interconnections.

Replace the damaged section of the harness.
Repair the engine harness. Refer to Procedure 019-043 in Section 19 (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html).
Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)</td><td>4A</td></tr>
</table>

STEP 2D. Check the add-on or accessory wiring at the positive (+) battery terminal.

Conditions :

- Turn keyswitch OFF.

Action	Specification/Repair	Next Step
Check the add-on or the accessory wiring at the positive (+) battery terminal. • Starting at the positive (+) battery terminal, follow any add-on or accessory wiring, and examine the wire(s) for damaged insulation or an installation error that can cause the supply wire to be shorted the engine block.	Damaged wires? **YES** **Repair :** Repair or replace the damaged wires. Refer to the OEM service manual.	**4A**
	Damaged wires? **NO**	**2E**

STEP 2E. Check the ignition input-to-ECM wire.

Conditions : • Turn keyswitch OFF.		
Action	**Specification/Repair**	**Next Step**
Check the ignition input-to-ECM wire. • Inspect the ignition wire from the ignition post in the ignition assembly to the ECM to make sure there are no interruptions in the wire, that is, no solenoids or relays.	Keyswitch wire uninterrupted? **YES**	**2F**
	Keyswitch wire uninterrupted? **NO** **Repair :** Correct the wiring so the wire is uninterrupted.	**4A**

STEP 2F. Check the ignition input circuit.

Conditions : • Turn keyswitch OFF. • Disconnect the engine harness connector from the ECM 50-pin connector.		
Action	**Specification/Repair**	**Next Step**

Check the ignition input circuit. • Measure the resistance from the ignition post in the ignition assembly to the ignition input SUPPLY pin of the 50-pin engine harness connector.	Less than 5 ohms? **YES**	**3A**
	Less than 5 ohms? **NO** **Repair :** Troubleshoot all harnesses connected in series to determine which contains the short circuit. Refer to the circuit diagram or wiring diagram for all harness interconnections. Replace the damaged section of the harness. • Repair the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) • Repair the OEM harness. Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html)	**4A**

STEP 3. Check the SAE J1939 data link 2 circuit.

STEP 3A. Inspect the engine harness connector pins.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness 60-pin connector from the ECM.

Action	Specification/Repair	Next Step

Inspect the engine harness connector and ECM connector pins for the following: • Loose connector • Corroded pins • Bent or broken pins • Pushed back or expanded pins • Moisture in or on the connector • Missing or damaged connector seals • Dirt or debris on or in the connector pins • Connector shell broken • Wire insulation damage • Damaged connector locking tab. Use the following procedure for general inspection techniques. Refer to Procedure 019-361 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)	Dirty or damaged pins? **YES** **Repair :** A damaged connection has been detected in the engine harness connector or ECM connector. • Flush the dirt, debris, or moisture from the connector pins. Use electrical contact cleaner, Part Number 3824510. • Install the appropriate connector seal if it is damaged or missing. Repair the damaged pins. Repair or replace the engine harness, OEM harness, or replace the ECM, whichever has the damaged pins. • Replace the damaged section of the engine harness or the ECM. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) • Replace the OEM harness. Refer to Procedure Refer to Procedure 019-071 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-071.html) • Replace the ECM. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/	**3D**

	procedures/105/105-019-031.html)	
	Dirty or damaged pins? **NO**	**3B**

STEP 3B. Check the SAE J1939 data link 2 engine network resistance in the engine harness.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60-pin connector.

Action	Specification/Repair	Next Step
Check the SAE J1939 data link 2 engine network resistance. • Measure the resistance between the SAE J1939 data link 2 engine network SUPPLY pin and the SAE J1939 data link 2 engine network RETURN pin in the ECM engine harness connector. • Refer to the circuit diagram or wiring diagram for connector pin for identification. • Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Between 54 to 66 ohms? **YES** **Repair :** The SAE J1939 data link 2 engine network resistance is within specification. There is a problem in the calibration. Check to make sure the appropriate calibrations are installed in the ECMs.	**3D**
	Between 54 to 66 ohms? **NO**	**3C**

STEP 3C. Check the SAE J1939 data link 2 engine network terminator resistor.

Conditions :

- Turn keyswitch OFF.
- Disconnect the engine harness connector from the ECM 60-pin connector.
- Disconnect one of the SAE J1939 data link 2 engine network terminator resistors from the engine harness.

Action	Specification/Repair	Next Step

Check the SAE J1939 data link 2 engine network resistance. Measure the resistance between the SAE J1939 data link 2 engine network SUPPLY pin and the SAE J1939 data link 2 engine network RETURN pin in the ECM engine harness connector. • Refer to the circuit diagram or wiring diagram for connector pin identification. • Use the following procedure for general resistance measurement techniques. Refer to Procedure 019-360 in Section 19. (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)	Between 108 to 132 ohms? **YES** **Repair :** The removed SAE J1939 data link 2 engine network terminator resistor is damaged. • Replace the SAE J1939 data link 2 engine network terminator resistor. Refer to Procedure 019-428 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-428.html)	**3D**

	Between 108 to 132 ohms? **NO** **Repair :** An open or shorted engine SAE J1939 data link 2 circuit or damaged terminator resistor still connected to the harness has been detected. Refer to the circuit diagram or wiring diagram for all harness interconnections. • Replace the damaged section of the engine harness. Refer to Procedure 019-043 in Section 19. (/qs3/pubsys2/xml/en/procedures/122/122-019-043.html) • Replace the teminator resistor. Refer to Procedure 019-428 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-428.html)	**3D**

STEP 3D. Check for ECM communication to INSITE™ electronic service tool.

Conditions :

- INSITE™ electronic service tool connected to the vehicle.
- Turn keyswitch ON.

Action	Specification/Repair	Next Step

Check for communication between INSITE™ electronic service tool and the ECM. Use INSITE™ electronic service tool to check for communication with the ECM.	Does the ECM communicate? **YES**	**4A**
	Does the ECM communicate? **NO**	**Reference the ECM - No Communication Troubleshooting Tree in Section TT.**

STEP 4. Clear the fault codes.

STEP 4A. Disable the fault code.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Disable the fault code. • Use INSITE™ electronic service tool to verify the fault code is inactive.	Fault Code 2727 inactive? **YES**	**4B**
	Fault Code 2727 inactive? **NO** **Repair :** Return to the troubleshooting steps or contact a Cummins® Authorized Repair Location if all steps have been completed and checked again.	**4B**

STEP 4B. Clear the inactive fault codes.

Conditions :

- Connect all components.
- Turn keyswitch ON.
- Connect INSITE™ electronic service tool.

Action	Specification/Repair	Next Step
Clear the inactive fault codes. • Use INSITE™ electronic service tool to clear the inactive fault codes.	All fault codes cleared? **YES**	**Repair complete**
	All fault codes cleared? **NO** **Repair :** Troubleshoot any remaining fault codes.	**Appropriate troubleshooting steps**

Last Modified: 18-Sep-2015